

UNIT → 1

Date

MATRICES

UNIT - 1

→ SQUARE MATRIX

Order $n \times n$

→ DIAGONAL MATRIX

A sq matrix with all non diagonal elements = 0

→ IDENTITY MATRIX

A sq matrix with all non diag = 0 & diag = 1

→ SINGULAR MATRIX

A sq matrix C such that $|C| = 0$

Q Show that if $AB = 0$ doesn't imply $A = 0$ or $B = 0$
Let us consider

$$A = \begin{bmatrix} -1 \\ 0 \end{bmatrix} \quad B = \begin{bmatrix} 0 & -1 \end{bmatrix}$$

$$AB = \begin{bmatrix} -1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & -1 \end{bmatrix} = \begin{bmatrix} 0 \end{bmatrix}$$

$AB = [0]$ but neither $A = 0$ nor $B = 0$

Q Show that matrix multiplication isn't commutative
consider

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$$

$$AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 3 \\ 0 & 7 \end{bmatrix}$$

$$BA = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 3 & 4 \end{bmatrix}$$

$AB \neq BA$

so it's not commutative

$$Q \quad A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad \text{adj}(A) = ?$$

$$M_{11} = 4$$

$$C_{11} = 4$$

cofactor Matrix C

$$M_{12} = 3$$

$$C_{12} = -3$$

$$M_{21} = 2$$

$$C_{21} = -2$$

$$C = \begin{bmatrix} 4 & -3 \\ -2 & 1 \end{bmatrix}$$

$$M_{22} = 1$$

$$C_{22} = +1$$

$$\text{adj}A = C^T = \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}$$

→ LET 'A' is a sq matrix of order 'n'

• IDEMPOTENT

$$A^2 = A$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

• INVOLUTARY

$$A^2 = I$$

• NILPOTENT

$$A^k = 0$$

; $k \leq n$ $k \rightarrow$ index

• SYMMETRIC

$$A^T = A$$

• SKEW SYMMETRIC

$$A^T = -A$$

• ORTHOGONAL

$$A A^T = I$$

• HERMITIAN

$$A^\theta = A$$

• SKEW HERMITIAN

$$A^\theta = -A$$

• UNITARY

$$A A^\theta = I$$

• NORMAL

$$A A^\theta = A^\theta A$$

θ is conjugated transpose

$$A^\theta = (\bar{A})^T$$

[sign chng then transpose]

→ LINEAR DEPENDENCE & INDEPENDENCE

• A set of vectors $v_1, v_2, v_3, \dots, v_n$ is called linearly independent if the only solution to $c_1 v_1 + c_2 v_2 + \dots + c_n v_n = \vec{0}$ is trivial ($c_i = 0$ i.e. $c_1 = c_2 = c_3 = \dots = c_n = 0$) & c_i are scalars

• If there exists a non trivial combination of given set of vectors $v_1, v_2, \dots, v_n$ (atleast one $c_i \neq 0$) that gives a $\vec{0}$ then vectors are linearly dependent.

Q: Let $v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ be two vectors in $\mathbb{R}^2$ show that they are linearly indep.

$$c_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow c_1 = c_2 = 0$$

$\Leftrightarrow$ L.I.D

v_1 & v_2 are linearly independent

Q $v_1 = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$ $v_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

$$c_1 \begin{bmatrix} 2 \\ 4 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 2c_1 + c_2 \\ 4c_1 + 2c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow 2C_1 + C_2 = 0$$

$$\Rightarrow C_2 = -2C_1 \quad \& \quad C_1 \rightarrow \text{arbitrary}$$

$$\left. \begin{array}{l} C_1 = 1 \quad C_2 = -2 \\ C_1 = 0 \quad C_2 = 0 \\ C_1 = -1 \quad C_2 = 2 \end{array} \right\} \text{non zero for } C_1 \& C_2$$

$\Rightarrow V_1 \& V_2$ are linearly dependent

$\rightarrow$ ROW ECHLEON FORM (REF)

A matrix is in REF if :-

- All non zero rows are above any row having all zero elements
- The leading entry (first non zero element from left of each non zero row) is to right of the leading entry of row above it
- All entries below each leading entry are zero.

$$\begin{array}{c} \uparrow \text{leading entry} \\ \left[\begin{array}{cccc} \textcircled{1} & 2 & 3 & 4 \\ 0 & \textcircled{1} & 5 & 6 \\ 0 & 0 & \textcircled{2} & 7 \\ 0 & 0 & 0 & 0 \end{array} \right] \rightarrow \text{It's in REF}$$

$\rightarrow$ REDUCED ROW ECHLEON FORM (RREF)

A matrix is in RREF if :-

- It's in REF
- Each leading entry is exactly 1
- Each leading entry which is 1 is the only non zero entry in its column.

$$\begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \checkmark$$

RREF ✓

$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 3 & 2 & 1 \\ 0 & 1 & 2 & 1 \end{bmatrix} \times$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \times$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \times$$

Q Reduce $\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 3 & 2 & 1 \\ 0 & 1 & 2 & 1 \end{bmatrix}$ in RREF

$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 3 & 2 & 1 \\ 0 & 1 & 2 & 1 \end{bmatrix} \xrightarrow{R_3 \leftrightarrow R_2 - 3R_3} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 3 & 2 & 1 \\ 0 & 0 & -4 & -2 \end{bmatrix} \rightarrow \text{it's now in REF}$$

$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 2/3 & 1/3 \\ 0 & 0 & 1 & 1/2 \end{bmatrix} \xleftarrow{R_4 \leftarrow -R_4} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 2/3 & 1/3 \\ 0 & 0 & -4 & -2 \end{bmatrix}$$

$$\downarrow R_2 \leftrightarrow R_2 - 2/3 R_3$$

$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1/2 \end{bmatrix} \xrightarrow{R_1 \rightarrow R_1 - R_3} \begin{bmatrix} 1 & 0 & 0 & -1/2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1/2 \end{bmatrix}$$

↓
it's in RREF Spiral

→ RANK OF A MATRIX

It is the number of non zero rows when a matrix is reduced in REF

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_2 - 2R_1} \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} \quad \text{Rank} = 1$$

$$\begin{bmatrix} 2 & 3 \\ 1 & 3 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow 2R_2 - R_1} \begin{bmatrix} 2 & 3 \\ 0 & 3 \end{bmatrix} \quad \text{Rank} = 2$$

→ GAUSS ELIMINATION METHOD

Goal of gauss elimination method is to transform a system of linear equations into a matrix of REF so we can solve eqⁿ using back substitution

• HOW TO APPLY

- Write augmented matrix of system of eq
- Write above matrix in REF
- Solve the system using back substitution (start from last eq and move upwards)

Q: SOLVE. $x + y + z = 6$

$$2y + 5z = -4$$

$$2x + 5y - z = 27$$

$$\begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 2 & 5 & -4 \\ 2 & 5 & -1 & 27 \end{bmatrix} \rightarrow \text{augmented matrix}$$

$$\begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 2 & 5 & -4 \\ 2 & 5 & -1 & 27 \end{bmatrix} \xrightarrow{R_3 \leftrightarrow R_3 - 2R_1} \begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 2 & 5 & -4 \\ 0 & 3 & -3 & 15 \end{bmatrix}$$

$$\downarrow R_3 \leftrightarrow R_3/3$$

$$\begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 2 & 5 & -4 \\ 0 & 0 & 7 & -14 \end{bmatrix} \xleftarrow{R_3 \leftrightarrow R_2 - 2R_3} \begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 2 & 5 & -4 \\ 0 & 1 & -1 & 5 \end{bmatrix}$$

REF ✓

$$7z = -14$$

$$\boxed{z = -2}$$

$$2y + 5z = -4$$

$$2y - 10 = -4$$

$$2y = 6$$

$$\boxed{y = 3}$$

$$x + y + z = 6$$

$$x + 3 - 2 = 6$$

$$\boxed{x = 5}$$

→ GAUSS JORDAN METHOD

- Same as gauss elimination
- write augmented matrix from eqⁿ
- Reduce it in RREF
- solve using back substitution

$$\begin{aligned} x + y + z &= 6 \\ 2x + 5z &= -4 \\ 2x + 5y - z &= 27 \end{aligned}$$

$$\begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 2 & 5 & -4 \\ 2 & 5 & -1 & 27 \end{bmatrix} \xrightarrow{\text{REF}} \begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 2 & 5 & -4 \\ 0 & 0 & 7 & -14 \end{bmatrix}$$

$$\begin{aligned} R_2 &\leftrightarrow R_2/2 \\ R_3 &\leftrightarrow R_3/7 \end{aligned}$$

$$\begin{bmatrix} 1 & 0 & -3/2 & 8 \\ 0 & 1 & +5/2 & -2 \\ 0 & 0 & 1 & -2 \end{bmatrix} \xleftarrow{R_1 \leftrightarrow R_1 - R_2} \begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & 5/2 & -2 \\ 0 & 0 & 1 & -2 \end{bmatrix}$$

$$\begin{aligned} R_2 &\leftrightarrow R_2 - 5/2 R_3 \\ R_1 &\leftrightarrow R_1 + \frac{3}{2} R_3 \end{aligned}$$

$$\begin{bmatrix} 1 & 0 & 0 & 8-3=5 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & -2 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & -2 \end{bmatrix}$$

$$z = -2$$

$$y = 3$$

$$x = 5$$

$$Q: \quad x + y + 2z = 2$$

$$3x + y + z = 1$$

$$2x + 2y + 4z = 4$$

$$\begin{bmatrix} 1 & 1 & 2 & 2 \\ 3 & 1 & 1 & 1 \\ 2 & 2 & 4 & 4 \end{bmatrix} \xrightarrow{R_3 \leftrightarrow R_3/2} \begin{bmatrix} 1 & 1 & 2 & 2 \\ 3 & 1 & 1 & 1 \\ 1 & 1 & 2 & 2 \end{bmatrix}$$

$$R_2 \leftrightarrow 3R_1 - R_2$$

$$\begin{bmatrix} 1 & 1 & 2 & 2 \\ 0 & 2 & 5 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xleftarrow{R_3 \leftrightarrow R_1 - R_3} \begin{bmatrix} 1 & 1 & 2 & 2 \\ 0 & 2 & 5 & 5 \\ 1 & 0 & 1 & 2 \end{bmatrix}$$

$$R_2 \leftrightarrow R_2/2$$

$$\begin{bmatrix} 1 & 1 & 2 & 2 \\ 0 & 1 & 5/2 & 5/2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 \rightarrow R_1 - R_2} \begin{bmatrix} 1 & 0 & -1/2 & -1/2 \\ 0 & 1 & 5/2 & 5/2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

RREF ✓

$$y + \frac{5}{2}z = \frac{5}{2}$$

$$x - \frac{1}{2}z = -\frac{1}{2}$$

$$y = \frac{5}{2}(1 - z)$$

$$x = \frac{1}{2}(z - 1)$$

 $z \rightarrow \text{arbitrary}$

- Finding inverse of a matrix using row operation
- Suppose we have a matrix A , first we have to form augmented matrix $[A/I]$ - I is identity matrix of same order
- Apply elementary row operations to transform the left side $^{(A)}$ of augmented matrix into the identity matrix
- When the left side becomes identity matrix then right side will be A^{-1}

Q Find inverse of $A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$ using elementary row ops

$$\left[\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 5 & 3 & 0 & 1 \end{array} \right] \xrightarrow{R_1 \rightarrow R_1/2} \left[\begin{array}{cc|cc} 1 & 1/2 & 1/2 & 0 \\ 5 & 3 & 0 & 1 \end{array} \right]$$

$$\downarrow R_2 \rightarrow R_2 - 5R_1$$

$$\left[\begin{array}{cc|cc} 1 & 0 & 3 & -1 \\ 0 & 1/2 & -5/2 & 1 \end{array} \right] \xleftarrow{R_1 \rightarrow R_1 - R_2} \left[\begin{array}{cc|cc} 1 & 1/2 & 3/2 & 0 \\ 0 & 1/2 & -5/2 & 1 \end{array} \right]$$

$$\downarrow R_2 \rightarrow 2R_2$$

$$\left[\begin{array}{cc|cc} 1 & 0 & 3 & -1 \\ 0 & 1 & -5 & 2 \end{array} \right]$$

$$A^{-1} = \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}$$

→ Cramer's Rule
It's a method for solving a system of linear eq using determinants, provided matrix is invertible or determinant of coeff matrix is non zero.

→ Cramer's Rule Formula

If determinant of coeff matrix is non zero then each variable is given by

$$x_i = \frac{\det(A_i)}{\det(A)}$$

$$i = 1, 2, \dots, n$$

$A_i \rightarrow$ matrix obtd from coeff matrix A by replacing the i^{th} column of A with column vector b

Q Solve $2x + y = 5$ using Cramer's Rule
 $x - y = 1$

$$A = \begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix}$$

Since $|A| = -3$

we can apply Cramer's rule

$$A_1 = \begin{bmatrix} 5 & 1 \\ 1 & -1 \end{bmatrix}$$

$$|A_1| = -6$$

$$\Rightarrow x = \frac{-6}{-3} = 2$$

$$A_2 = \begin{bmatrix} 2 & 5 \\ 1 & 1 \end{bmatrix}$$

$$|A_2| = -3$$

$$\Rightarrow y = \frac{-3}{-3} = 1$$

$$x = 2 \quad \& \quad y = 1$$

Q Test the linear independence

$$v_1 = [1, 1, 0]$$

$$v_2 = [1, 0, 1]$$

$$v_3 = [0, 1, 1]$$

$$c_1 + c_2 = 0$$

$$c_1 = -c_2$$

$$c_1 + c_3 = 0$$

$$c_3 = -c_1$$

$$c_2 + c_3 = 0$$

$$c_3 = 0$$

$$2c_2 = 0$$

$$c_1 = 0$$

$$c_2 = 0$$

all $c_i = 0$ hence linearly independent

Q Solve the system of linear eq

$$3x + 2y + z = 1$$

$$2x - 2y + 4z = -2$$

$$-x + 1/2 y - z = 0$$

$$\left[\begin{array}{ccc|c} 3 & 2 & -1 & 1 \\ 2 & -2 & 4 & -2 \\ -1 & 1/2 & -1 & 0 \end{array} \right]$$

$$\downarrow R_2 \rightarrow \frac{R_2}{2}$$

$$\left[\begin{array}{ccc|c} 3 & 2 & -1 & 1 \\ 2 & -2 & 4 & -2 \\ -1 & 1/2 & -1 & 0 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2/2} \left[\begin{array}{ccc|c} 3 & 2 & -1 & 1 \\ 1 & -1 & 2 & -1 \\ -1 & 1/2 & -1 & 0 \end{array} \right] \xrightarrow{R_3 \rightarrow R_1 + R_3} \left[\begin{array}{ccc|c} 3 & 2 & -1 & 1 \\ 1 & -1 & 2 & -1 \\ 0 & 1/2 & 1 & -1 \end{array} \right]$$

$$\begin{bmatrix} 3 & 2 & -1 & 1 \\ 2 & -2 & 4 & -2 \\ -1 & 1/2 & -1 & 0 \end{bmatrix} \xrightarrow[\substack{R_3 \rightarrow 2R_3 \\ R_2 \rightarrow \frac{R_2}{2}}]{R_3 \rightarrow 2R_3} \begin{bmatrix} 3 & 2 & -1 & 1 \\ 1 & -1 & 2 & -1 \\ -2 & 1 & -2 & 0 \end{bmatrix}$$

$$\downarrow R_1 \rightarrow R_1/3$$

$$\begin{bmatrix} 1 & 2/3 & -1/3 & 1/3 \\ 0 & 5/3 & -7/3 & 4/3 \\ 0 & 7/3 & -8/3 & 2/3 \end{bmatrix} \xrightarrow[\substack{R_3 \rightarrow R_3 + 2R_1 \\ R_2 \rightarrow R_1 - R_2}]{R_2 \rightarrow R_1 - R_2} \begin{bmatrix} 1 & 2/3 & -1/3 & 1/3 \\ 1 & -1 & 2 & -1 \\ 2 & 1 & -2 & 0 \end{bmatrix}$$

$$\downarrow R_2 \rightarrow \frac{3}{5} R_2$$

$$\begin{bmatrix} 1 & 2/3 & -1/3 & 1/3 \\ 0 & 1 & -7/5 & 4/5 \\ 0 & 7/3 & -8/3 & 2/3 \end{bmatrix} \xrightarrow[\substack{R_3 \rightarrow R_3 - \frac{7}{3} R_2 \\ R_1 \rightarrow R_1 - \frac{2}{3} R_2}]{R_1 \rightarrow R_1 - \frac{2}{3} R_2} \begin{bmatrix} 1 & 0 & 3/5 & -1/5 \\ 0 & 1 & -7/5 & 4/5 \\ 0 & 0 & 3/5 & -6/5 \end{bmatrix}$$

$$\downarrow R_3 \rightarrow \frac{5}{3} R_3$$

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & -2 \end{bmatrix} \xrightarrow[\substack{R_1 \rightarrow R_1 - \frac{3}{5} R_3 \\ R_2 \rightarrow R_2 + \frac{7}{5} R_3}]{R_2 \rightarrow R_2 + \frac{7}{5} R_3} \begin{bmatrix} 1 & 0 & 3/5 & -1/5 \\ 0 & 1 & -7/5 & 4/5 \\ 0 & 0 & 1 & -2 \end{bmatrix}$$

$$z = -2$$

$$y = -2$$

$$x = 1$$

→ consistent & inconsistent linear system of eq
 A system of linear eq is said to be consistent if it has atleast 1 solution (unique / ∞ solⁿ)

A system of linear eq is said to be inconsistent if it has no solution.

→ Result Suppose $A_{n \times n} X_{n \times 1} = B_{n \times 1}$

If $\text{Rank}(A) = \text{Rank}([A/B]) = n$ (unique solⁿ)

If $\text{Rank}(A) = \text{Rank}([A/B]) < n$ (∞ solⁿ)

If $\text{Rank}(A) \neq \text{Rank}([A/B])$ (no solⁿ)

Q Check consistent / inconsistent & comment about no of solⁿ

$$3x + 2y - z = 1$$

$$2x - 2y + 4z = -2$$

$$-x + \frac{1}{2}y - z = 0$$

$$A = \begin{bmatrix} 3 & 2 & -1 \\ 2 & -2 & 4 \\ -1 & 1/2 & -1 \end{bmatrix} \quad B = \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix}$$

$$A/B = \left[\begin{array}{ccc|c} 3 & 2 & -1 & 1 \\ 2 & -2 & 4 & -2 \\ -1 & 1/2 & -1 & 0 \end{array} \right] \xrightarrow[R_3 \rightarrow 2R_3]{R_2 \rightarrow R_2/2} \left[\begin{array}{ccc|c} 3 & 2 & -1 & 1 \\ 1 & -1 & 2 & -1 \\ -2 & 1 & -2 & 0 \end{array} \right]$$

Date

$$\begin{bmatrix} 3 & 2 & -1 & 1 \\ 1 & -1 & 2 & -1 \\ -2 & 1 & -2 & 0 \end{bmatrix} \xrightarrow[R_3 \rightarrow R_3 + 2R_1]{R_2 \rightarrow 3R_2 - R_1} \begin{bmatrix} 3 & 2 & -1 & 1 \\ 0 & -5 & 7 & -4 \\ 0 & 5 & -4 & 2 \end{bmatrix}$$

$$R_3 \rightarrow R_2 + R_3$$

$$\text{Rank}(A) = 3$$

$$\text{Rank}(A/B) = 3$$

Consistent (Unique Sol)

$$\begin{bmatrix} 3 & 2 & -1 & 1 \\ 0 & -5 & 7 & -4 \\ 0 & 0 & 3 & -2 \end{bmatrix}$$

$$Q \quad 3x + 2y = 2$$

$$x + 2y = 2 \quad 0 = 1 - (A - S)$$

$$\begin{bmatrix} 3 & 2 & 2 \\ 1 & 2 & 2 \end{bmatrix} \xrightarrow{R_2 \rightarrow 3R_2 - R_1} \begin{bmatrix} 3 & 2 & 2 \\ 0 & 4 & 4 \end{bmatrix}$$

$$R_A = 2$$

$$R_{A/B} = 2$$

Consistent (Unique)

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$0 = E + NP - I$$

$$0 = E + NP - I$$

$$0 = (E - I) + NP$$

$$E = I - NP$$

Spiral